

Hamza Madi

Computer Science and IT

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EDUCATION	Feb 2022- Dec 2025	University Of KwaZulu-Natal, Durban BSc in Computer Science and IT Computer Science graduate and Artificial Intelligent intern with experience in Python, data analysis, and full-stack development. Built database-driven applications and supported students as a university demonstrator, strengthening both technical and communication skills.
		Relevant Coursework Computer Science: Data Structures, Algorithms, Programming Languages, Web Programming, Machine Learning Informatics: Computer Networks, Databases, Client-Side Development, Systems Design and Analysis
	Feb 2026- Progress	IBM Data Engineering Professional Certificate (In Progress) — IBM Learning data pipelines, ETL processes, SQL, Python, and cloud-based data engineering
PROFESSIONAL EXPERIENCE	Jan 2026- Present	AI and Data Science intern – Gijima • Gained hands-on experience in artificial intelligence and data science concepts, including data processing, analysis, and model development. • Worked with structured and semi-structured datasets to extract insights and support decision-making. • Applied Python for data analysis and basic machine learning tasks. • Collaborated with team members on real-world data-driven projects and problem-solving tasks. • Developed an understanding of data pipelines, data cleaning, and visualization techniques.
	July 2025- Nov 2025	Computer Systems Demonstrator, University of Kwazulu-Natal • Led demonstrations on Unix/Linux systems and computer networking concepts, including network protocols, routing, and system performance. • Implemented data distributions to analyse network load and resource utilization. • Provided hands-on support for network simulations and real-time traffic analysis.
	July 2024- Nov 2024	Java Programming Demonstrator, University of KwaZulu-Natal • Supporting first-year students with Object-Oriented Programming (OOP) and Java. • Delivering explanations of complex concepts, troubleshooting code, and guiding hands-on programming exercises. • Strengthened teaching, communication, and foundational software development skills.

Personal Projects

Manufacturing Defect Prediction System:

- Built a machine learning model to predict defective products using manufacturing data (cost, machine downtime, productivity, energy usage).
- Performed data cleaning, feature engineering, and applied classification algorithms to identify key factors contributing to defects.

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Power BI

Clinic Management System Web Application :

- Designed and implemented clinic management solution for patient records, and appointments.
- Created database-driven architecture with reporting dashboards

Technologies: .NET (C#), MySQL, HTML, CSS, JavaScript, Power BI

ProNearby Platform:

- Built location-based service platform connecting users with nearby professionals
- Developed backend using Flask with PostgreSQL database integration
- Implemented map-based functionality using Mapbox API

Technologies: Python, Flask, PostgreSQL, JavaScript, Mapbox

MzansiBuilds – Developer Collaboration Platform:

- Designed and developed a full-stack platform for developers to create projects, share progress, and collaborate in real time.
- Implemented authentication, project management, collaboration requests, project feeds, and a celebration wall for completed projects.

Technologies: React.js, Flask (Python), PostgreSQL, JavaScript, HTML, CSS

- Explore my projects on **GitHub**:
[github.com/lionhamza](<https://github.com/lionhamza>)

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)

TECHNICAL SKILLS

- ❖ **Programming Languages:** Java, Python, C++, C#, SQL.
- ❖ **Databases:** PostgreSQL, MySQL
- ❖ **Backend & Web:** Flask, REST APIs, HTML, CSS, JavaScript
- ❖ **Data & Analysis Tools:** Debugging, testing strategies, full software development lifecycle. MS Excel (Advanced Formulas, Pivot Tables), Power BI

SKILLS & ABILITIES

- Strong problem-solving and debugging skills with attention to detail.
 - Experienced in building responsive, scalable, and user-focused applications.
 - Effective communicator and collaborator in cross-functional teams.
 - Quick learner, adaptable to new tools, frameworks, and emerging technologies.
 - Skilled in the full software development lifecycle, from design to deployment and software testing.
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